Adv ML for Gen-AI

Gaussian Mixture Model

https://ml-graph.github.io/spring-2026/

Yu Wang, Ph.D.
Assistant Professor
Department of Computer Science
University of Oregon
Personal: https://yuwang0103.github.io/
Lab: https://kindlab-fly.github.io/

Data Distribution

Dog

Cat

Data Distribution

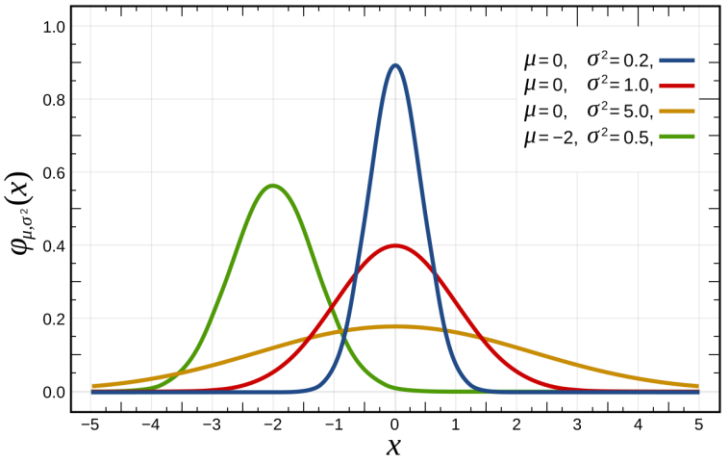
Dog – P(Dog)

Cat – P(Cat)

1. **There is no concrete image/shape of the dog, everyone can come up with one of your own choice**

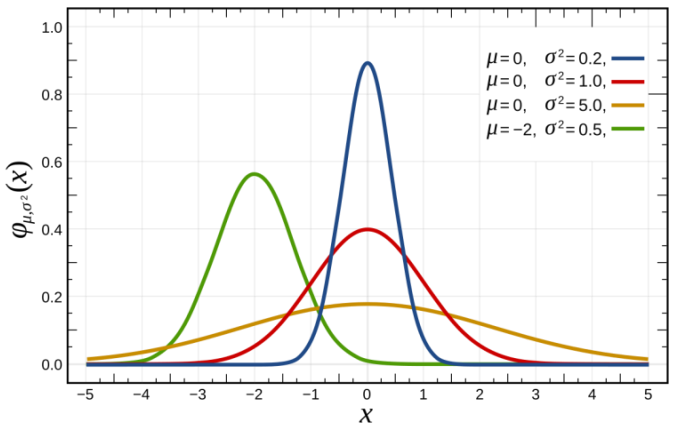
2. **But somehow dog and cat image distributions are different**

When you draw an image, you are actually sampling from a probability distribution!



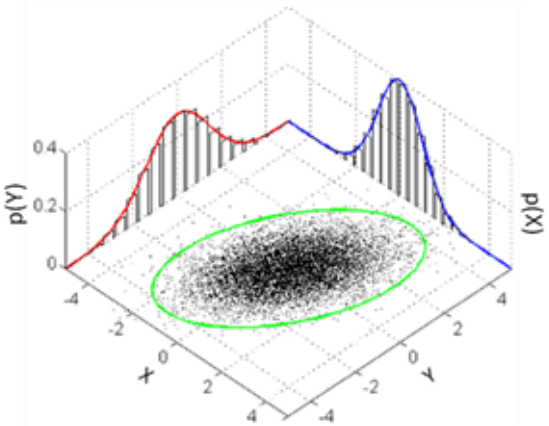
Data Distribution

1D Gaussian Distribution



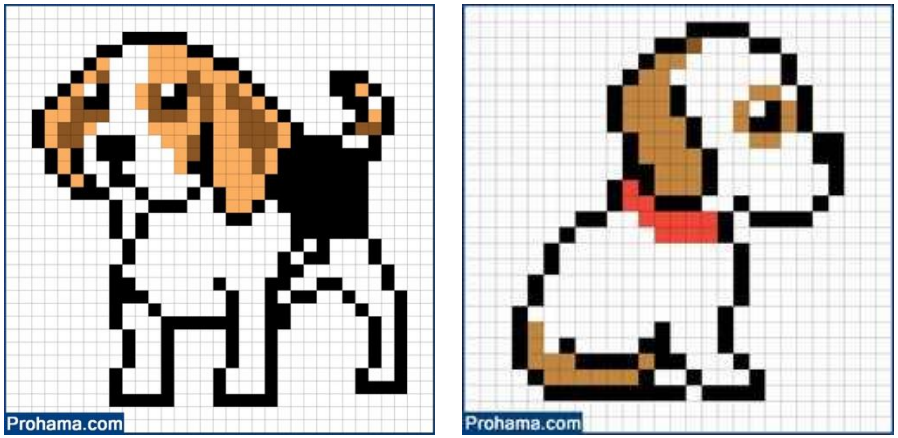
$\mathbb{R}$

2D Gaussian Distribution



$\mathbb{R}^2$

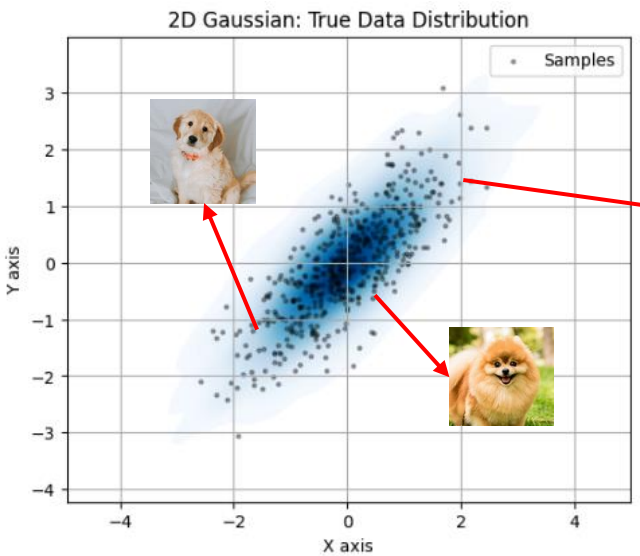
$\mathbb{R}^{256\times256}$



$\mathbb{R}^{256\times256}$

Estimating Data Distribution and Sampling from it

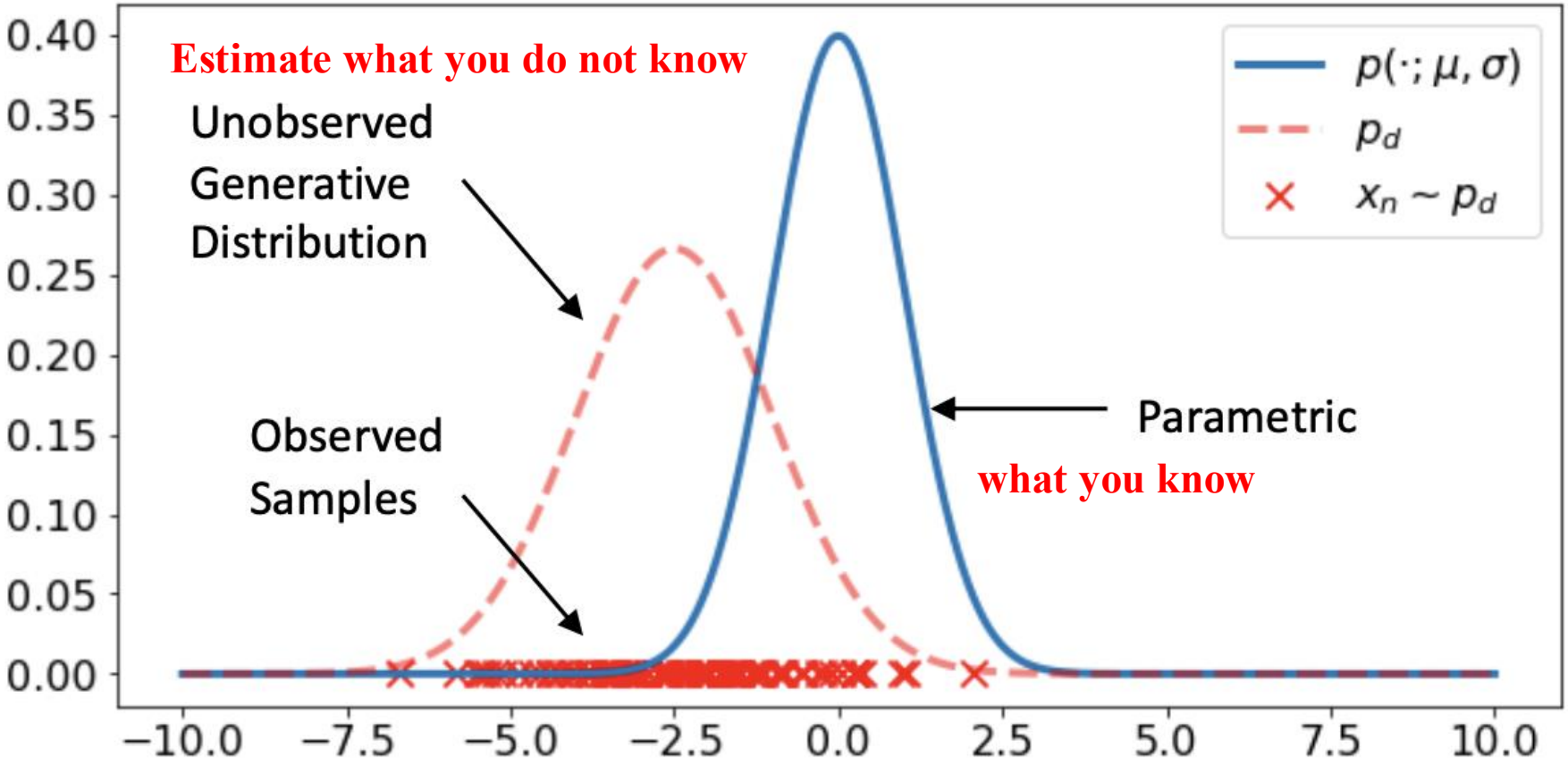
Estimate Data Distribution



Sampling from it

Gaussian Kernel Density Estimation

Using **existing function** to **estimate what you do not know** that **can best fit your observation**



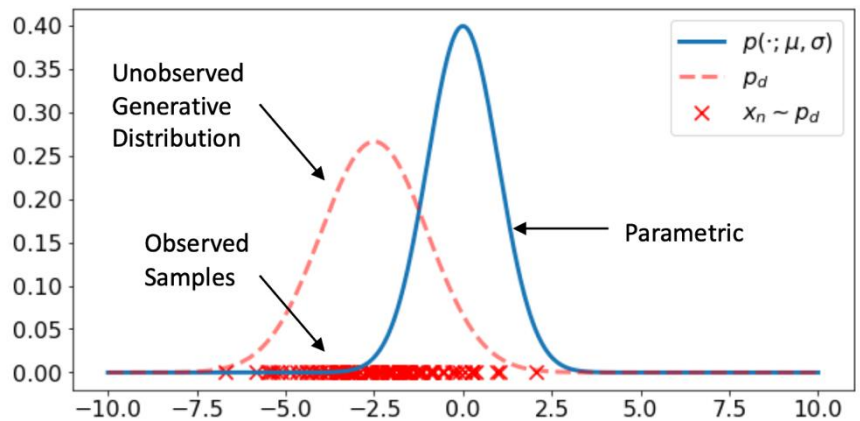
Gaussian Kernel Density Estimation

Using existing function to estimate what you do not know that can best fit your observation

$$\{x_i | x_i \sim P_d\}_{i=1}^{N}$$

$$p(x;\mu,\sigma) = N(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

What is μ, σ ?



1- Maximum Likelihood:

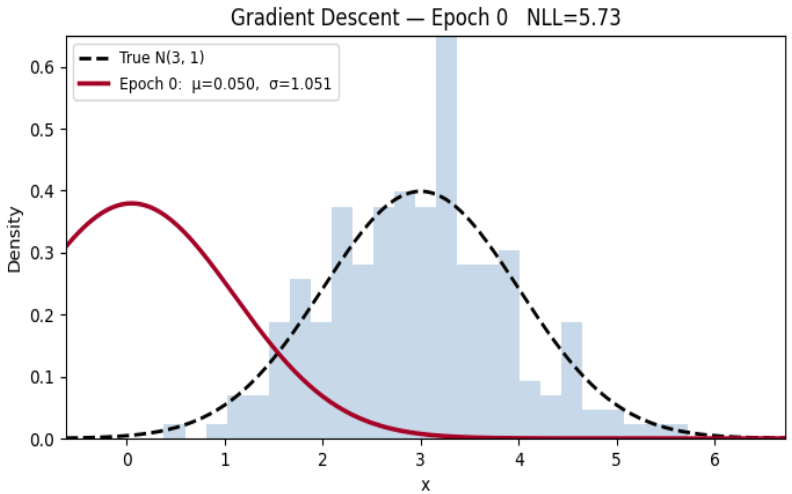
$$\underset{\mu,\sigma}{\operatorname{argmax}}\, p(x_1, \dots, x_N; \mu, \sigma) = \prod_{n=1}^{N} p(x_n; \mu, \sigma)$$

Gaussian Kernal Density for 1D data

```python
import torch
import numpy as np

x = torch.tensor(data, dtype=torch.float32)

# — Initialization ————————————————————————————————————————
mu       = torch.tensor(0.0, requires_grad=True)   # bad starting guess
log_sigma = torch.tensor(0.0, requires_grad=True)  # optimise log σ for stability

optimizer = torch.optim.Adam([mu, log_sigma], lr=0.05)

n_epochs     = 300
record_every = 30
history      = []   # (epoch, mu, sigma, nll)

# — Training loop ————————————————————————————————————————
for epoch in range(n_epochs):
    sigma = torch.exp(log_sigma)
    nll   = (
        0.5 * torch.log(2 * torch.tensor(np.pi))
        + log_sigma
        + 0.5 * ((x - mu) ** 2 / sigma ** 2).mean()
    )
    optimizer.zero_grad()
    nll.backward()
    optimizer.step()

    if epoch % record_every == 0 or epoch == n_epochs - 1:
        history.append((epoch, mu.item(), torch.exp(log_sigma).item(), nll.item()))

# — Training log ————————————————————————————————————————
print(f"{'Epoch':>6}  {'μ (mu)':>10}  {'σ (sigma)':>10}  {'NLL':>10}")
print("-" * 44)
for epoch, mu_val, sigma_val, nll_val in history:
    print(f"{epoch:>6}  {mu_val:>10.4f}  {sigma_val:>10.4f}  {nll_val:>10.4f}")

print(f"\nLearned  μ = {history[-1][1]:.4f},  σ = {history[-1][2]:.4f}")
print(f"True     μ = 3.0000,  σ = 1.0000")
```

Epoch	μ (mu)	σ (sigma)	NLL
0	0.0500	1.0513	5.7287
30	1.1211	2.4646	2.1749
60	1.6300	2.1697	1.9816
90	2.1959	1.5625	1.6727
120	2.8080	1.0914	1.3831
150	2.9932	0.9118	1.3460
180	2.9524	0.9273	1.3450
210	2.9605	0.9297	1.3449
240	2.9590	0.9284	1.3449
270	2.9593	0.9287	1.3449
299	2.9592	0.9287	1.3449

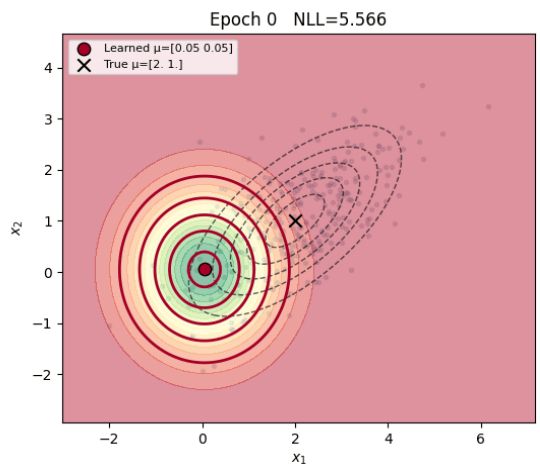
Learned μ = 2.9592, σ = 0.9287
True μ = 3.0000, σ = 1.0000



Gaussian Kernal Density for 2D data

```python
import torch

X2d = torch.tensor(data_2d, dtype=torch.float32)   # (N, 2)

# ── Initialization — bad spherical guess ──────────────────────────────
mu2d = torch.tensor([0.0, 0.0], requires_grad=True)

# Cholesky factor L: lower-triangular with positive diagonal (via log)
L_diag = torch.tensor([0.0, 0.0], requires_grad=True)   # log of diagonal entries
L_off  = torch.tensor([0.0],      requires_grad=True)   # single off-diagonal entry

def build_L():
    """Reconstruct lower-triangular Cholesky factor."""
    L = torch.zeros(2, 2)
    L[0, 0] = torch.exp(L_diag[0])
    L[1, 1] = torch.exp(L_diag[1])
    L[1, 0] = L_off[0]
    return L

optimizer2d = torch.optim.Adam([mu2d, L_diag, L_off], lr=0.05)

n_epochs2d     = 400
record_every2d = 40
history2d      = []   # (epoch, mu_np, cov_np, nll)

# ── Training loop ─────────────────────────────────────────────────────
for epoch in range(n_epochs2d):
    L   = build_L()
    Sig = L @ L.t()                        # Σ = L Lᵀ  (always PSD)
    Sig_inv = torch.linalg.inv(Sig)
    log_det = 2 * L_diag.sum()             # log|Σ| = 2 * sum(log diag L)

    diff = X2d - mu2d                      # (N, 2)
    mahal = (diff @ Sig_inv * diff).sum(dim=1)  # (N,)
    nll2d = 0.5 * (2 * torch.log(torch.tensor(2 * torch.pi)) + log_det + mahal).mean()

    optimizer2d.zero_grad()
    nll2d.backward()
    optimizer2d.step()

    if epoch % record_every2d == 0 or epoch == n_epochs2d - 1:
        history2d.append((
            epoch,
            mu2d.detach().numpy().copy(),
            Sig.detach().numpy().copy(),
            nll2d.item()
        ))
```

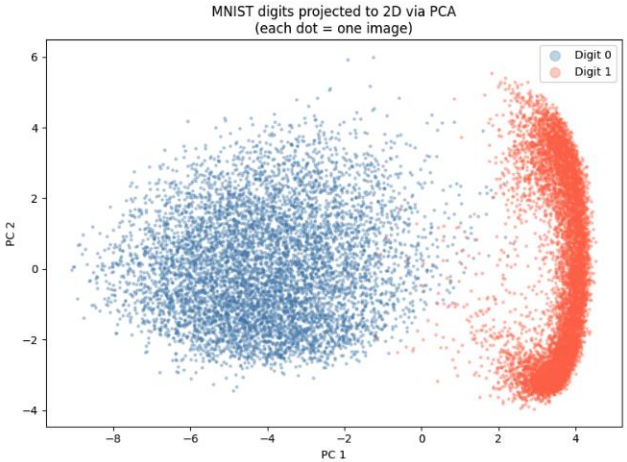
```
 Epoch                μ            Σ diag        NLL
-----------------------------------------------------------
     0        [0.05 0.05]          [1. 1.]     5.5661
    40      [1.344 0.587]     [3.043 1.454]     2.8954
    80      [1.939 0.958]     [1.604 0.971]     2.7095
   120      [2.031 1.008]     [1.435 0.931]     2.7043
   160      [2.017 1.   ]     [1.441 0.938]     2.7042
   200      [2.019 1.001]     [1.442 0.938]     2.7042
   240      [2.019 1.001]     [1.441 0.938]     2.7042
   280      [2.019 1.001]     [1.442 0.938]     2.7042
   320      [2.019 1.001]     [1.442 0.938]     2.7042
   360      [2.019 1.001]     [1.442 0.938]     2.7042
   399      [2.019 1.001]     [1.442 0.938]     2.7042

Learned  μ = [2.019 1.001]
True     μ = [2. 1.]

Learned  Σ =
[[1.442 0.766]
 [0.766 0.938]]
True     Σ =
[[1.5 0.8]
 [0.8 1. ]]
```



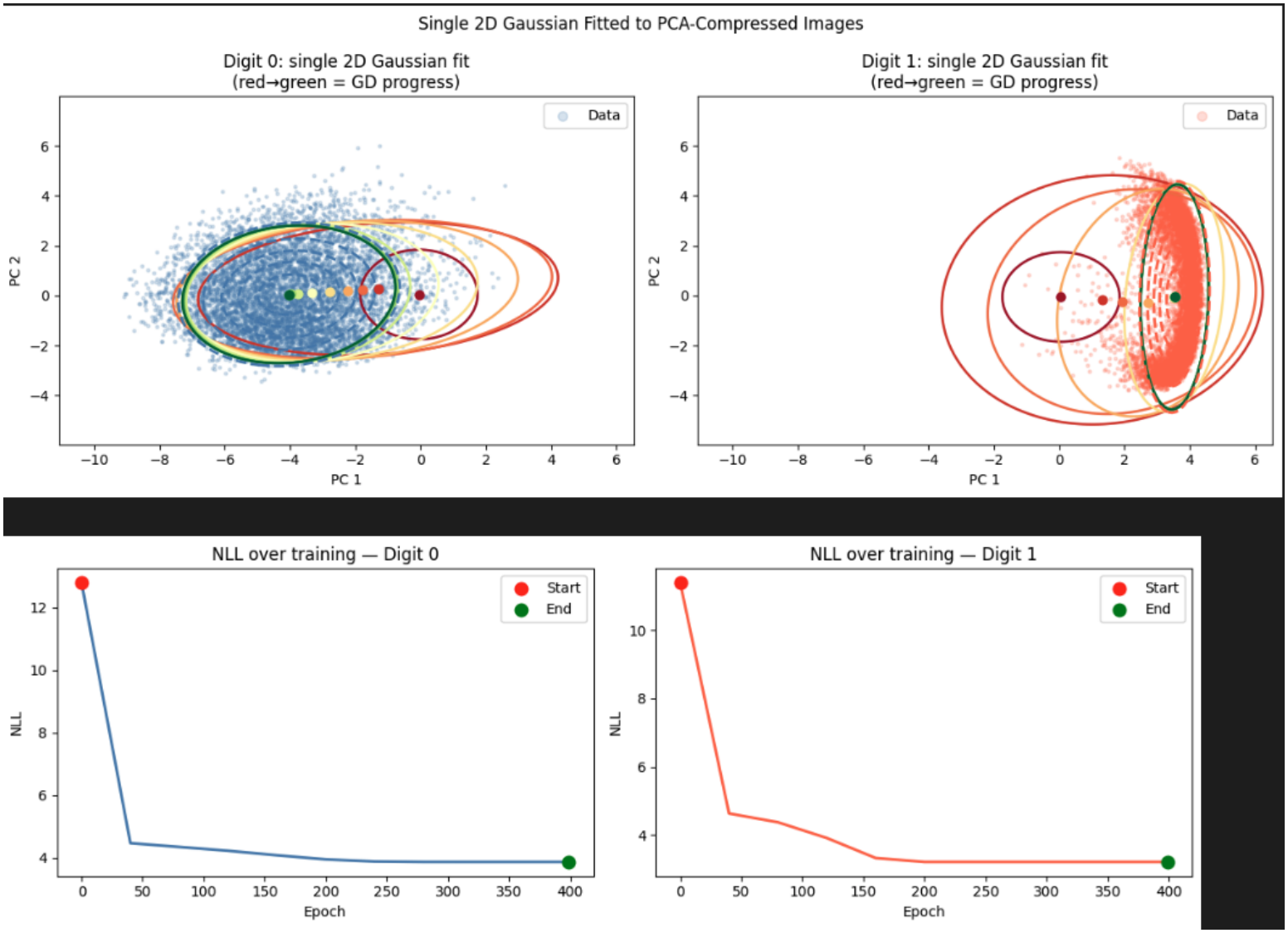
Gaussian Kernal Density for ND data

Dataset: 14780 images (6903 zeros, 7877 ones)

```python
# — Fit PCA: 784-D → 2-D ————————————————————————————————
pca = PCA(n_components=2, random_state=42)
X_2d = pca.fit_transform(X_raw)   # shape (N, 2)

print(f'Explained variance: PC1={pca.explained_variance_ratio_[0]:.1%},  '
      f'PC2={pca.explained_variance_ratio_[1]:.1%}')
print(f'Total captured: {pca.explained_variance_ratio_.sum():.1%} of variation')

# — Scatter plot of the 2D PCA space ————————————————————
fig, ax = plt.subplots(figsize=(8, 6))
colors = {0: 'steelblue', 1: 'tomato'}
for digit in [0, 1]:
    idx = y_raw == digit
    ax.scatter(X_2d[idx, 0], X_2d[idx, 1],
               c=colors[digit], alpha=0.3, s=5, label=f'Digit {digit}')

ax.set_title('MNIST digits projected to 2D via PCA\n(each dot = one image)')
ax.set_xlabel('PC 1')
ax.set_ylabel('PC 2')
ax.legend(markerscale=4)
plt.tight_layout()
plt.show()
```



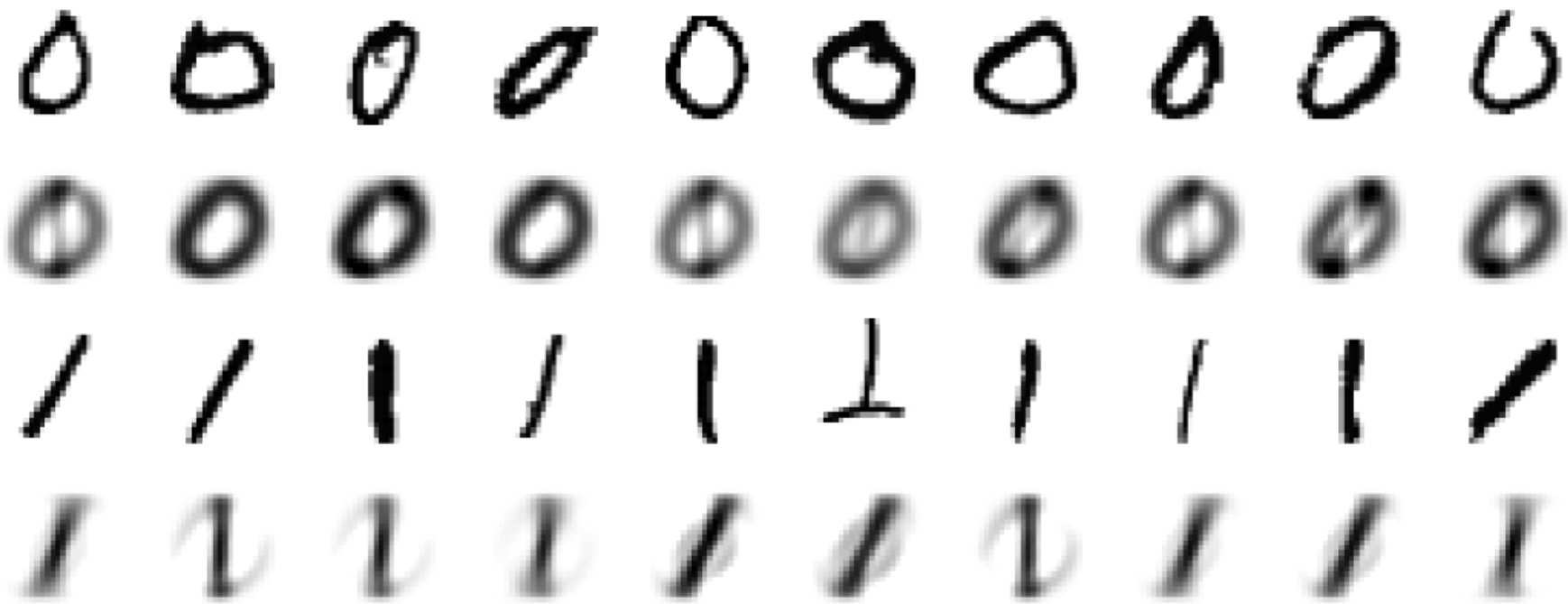
Gaussian Kernal Density for ND data



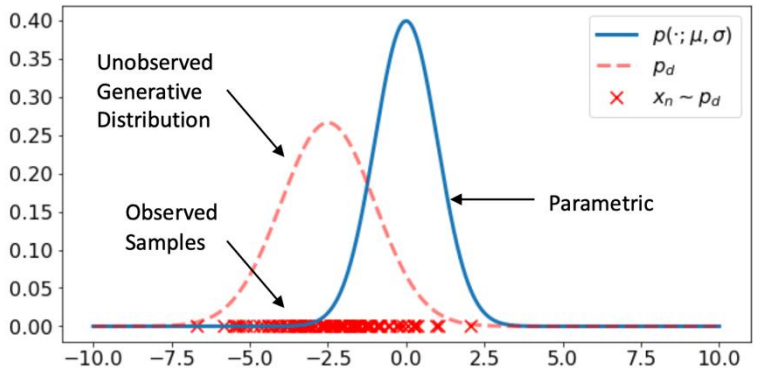
Gaussian Kernal Density for ND data

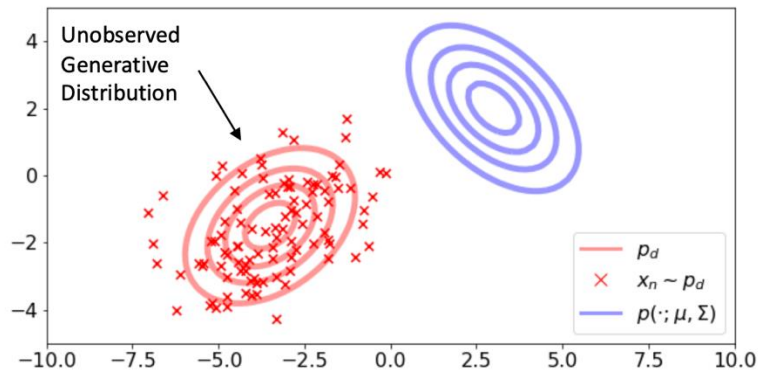
```python
# — Generated images via KDE sampling ————————————————————————
gen_2d = kdes[digit].resample(n_generate, seed=digit).T   # (n_generate, 2)
gen_images = pca.inverse_transform(gen_2d)                # (n_generate, 784)
gen_images = np.clip(gen_images, 0, 1)
```

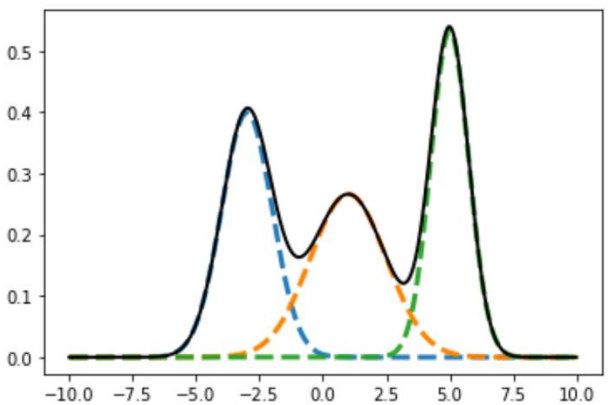
Real vs. KDE-Generated Images

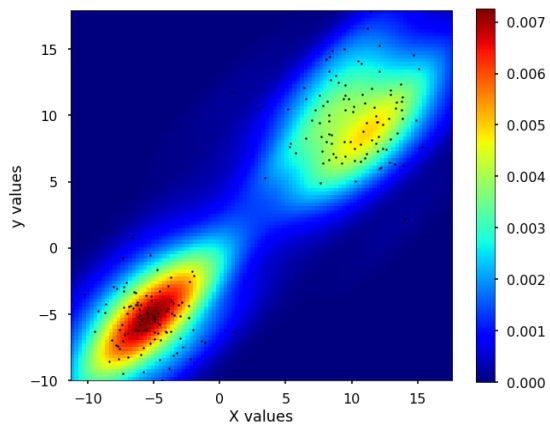


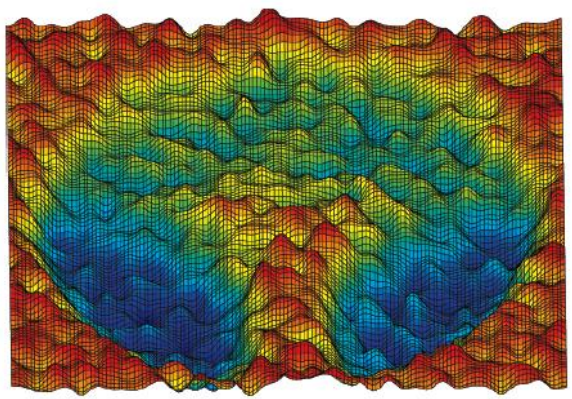
Problem with only using 1 Gaussian Kernal Density





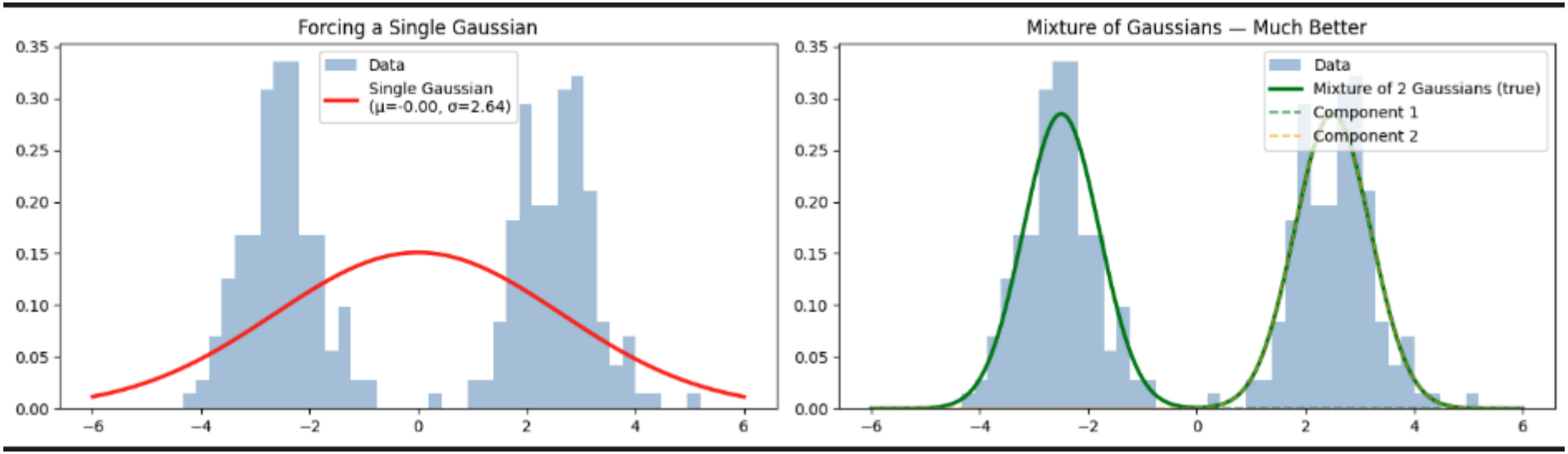






If using one existing function to estimate is not enough, then let's try more!

Gaussian Mixture Models



$$p(x) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(x \mid \mu_k, \sigma_k)$$

Symbol	Name	Meaning
π_k	**mixing weight**	How often this component is chosen. Must satisfy $\pi_k \geq 0$ and $\sum_k \pi_k = 1$.
μ_k	**mean**	Where the k-th Gaussian is centered.
σ_k	**standard deviation**	How wide the k-th Gaussian is.

Gaussian Mixture Models

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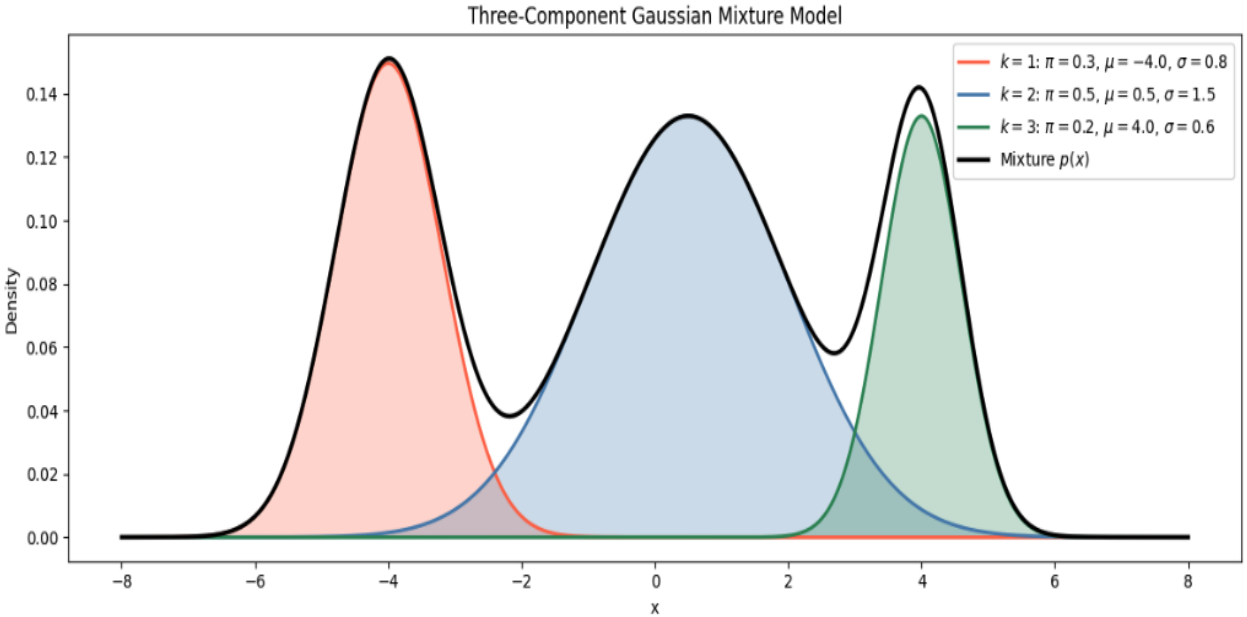
```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

# --- Visualise the GMM formula: components + mixture ---
x_plot = np.linspace(-8, 8, 500)

# Three-component GMM
pis    = [0.3,  0.5,  0.2]
mus    = [-4.0,  0.5,  4.0]
sigmas = [ 0.8,  1.5,  0.6]
colors = ['tomato', 'steelblue', 'seagreen']

mixture = np.zeros_like(x_plot)
fig, ax = plt.subplots(figsize=(12, 5))

for k, (pi, mu, sigma, color) in enumerate(zip(pis, mus, sigmas, colors)):
    component = pi * norm.pdf(x_plot, mu, sigma)
    mixture  += component
    ax.fill_between(x_plot, component, alpha=0.25, color=color)
    ax.plot(x_plot, component, color=color, lw=2,
            label=fr'$k={k+1}$: $\pi={pi}$, $\mu={mu}$, $\sigma={sigma}$')

ax.plot(x_plot, mixture, 'k-', lw=2.5, label='Mixture $p(x)$')
ax.set_xlabel('x')
ax.set_ylabel('Density')
ax.set_title('Three-Component Gaussian Mixture Model')
ax.legend()
plt.tight_layout()
plt.show()
```



Sampling from Gaussian Mixture Models

$$p(x) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(x \mid \mu_k, \sigma_k) \qquad \sum_k \pi_k = 1.$$

Symbol	Name	Meaning
π_k	**mixing weight**	How often this component is chosen. Must satisfy $\pi_k \geq 0$ and $\sum_k \pi_k = 1$.
μ_k	**mean**	Where the k-th Gaussian is centered.
σ_k	**standard deviation**	How wide the k-th Gaussian is.

Because the GMM has a clean generative story, sampling is easy:

1. Draw a component index $z \in \{1, \ldots, K\}$ with probabilities $[\pi_1, \ldots, \pi_K]$.
2. Draw $x \sim \mathcal{N}(\mu_z, \sigma_z)$.

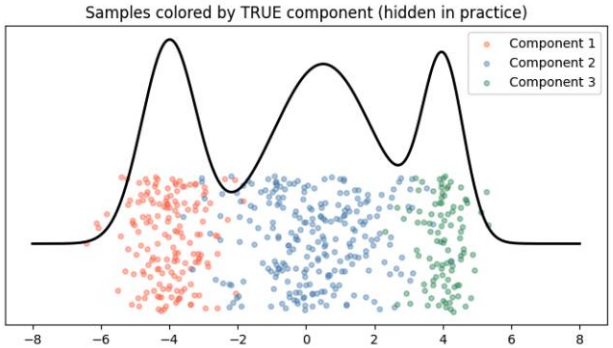
Sampling from Gaussian Mixture Models

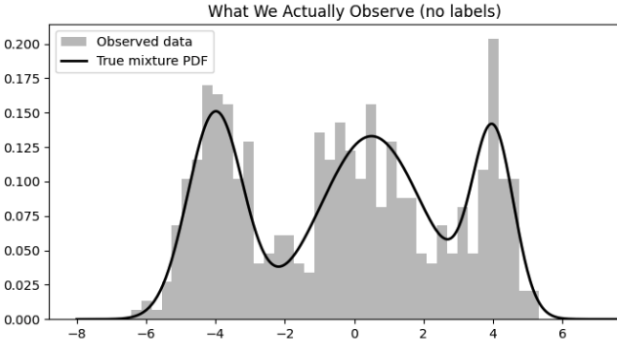
$$p(x) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(x \mid \mu_k, \sigma_k)$$

Because the GMM has a clean generative story, sampling is easy:

1. Draw a component index $z \in \{1, \ldots, K\}$ with probabilities $[\pi_1, \ldots, \pi_K]$.
2. Draw $x \sim \mathcal{N}(\mu_z, \sigma_z)$.

```python
np.random.seed(0)

# True GMM parameters
pis_true    = np.array([0.3, 0.5, 0.2])
mus_true    = np.array([-4.0, 0.5, 4.0])
sigmas_true = np.array([0.8, 1.5, 0.6])
K = len(pis_true)
N = 500

# Step 1: draw component labels
z = np.random.choice(K, size=N, p=pis_true)      # shape (N,)

# Step 2: draw samples from assigned components
x_samples = np.array([np.random.normal(mus_true[zi], sigmas_true[zi]) for zi in z])

x_plot = np.linspace(-8, 8, 400)
mixture_pdf = sum(pi * norm.pdf(x_plot, mu, sigma)
                  for pi, mu, sigma in zip(pis_true, mus_true, sigmas_true))

fig, axes = plt.subplots(1, 2, figsize=(14, 4))

# Left: colored by true component (normally hidden!)
for k in range(K):
    axes[0].scatter(x_samples[z == k], np.zeros(np.sum(z == k)) + np.random.uniform(-0.05, 0.05, np.sum(z == k)),
                    alpha=0.4, s=15, color=colors[k], label=f'Component {k+1}')
axes[0].plot(x_plot, mixture_pdf, 'k-', lw=2)
axes[0].set_yticks([])
axes[0].set_title('Samples colored by TRUE component (hidden in practice)')
axes[0].legend()

# Right: what we actually observe — unlabeled
axes[1].hist(x_samples, bins=40, density=True, alpha=0.5, color='gray', label='Observed data')
axes[1].plot(x_plot, mixture_pdf, 'k-', lw=2, label='True mixture PDF')
axes[1].set_title('What We Actually Observe (no labels)')
axes[1].legend()

plt.tight_layout()
plt.show()

print(f'Samples per component (true): {[(z==k).sum() for k in range(K)]}')
```

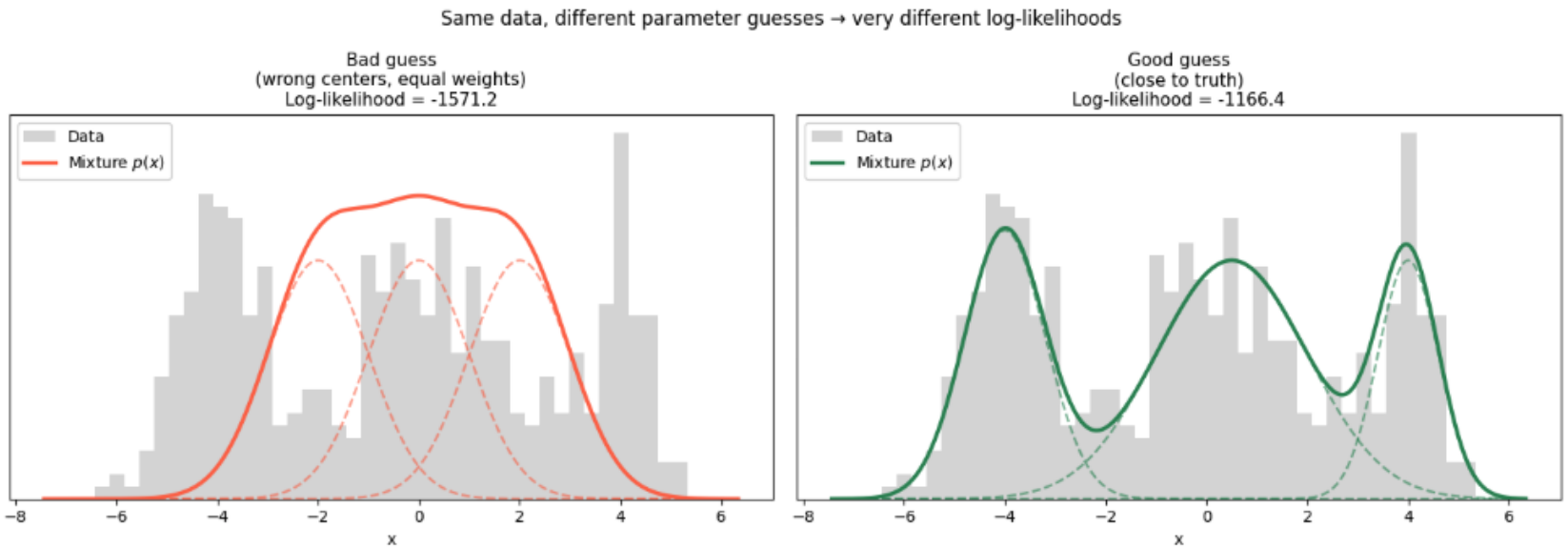




Given Observed Samples, Good Guess, Bad Guess

$$\log \mathcal{L}(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\sigma}) = \sum_{i=1}^{N} \log \left(\sum_{k=1}^{K} \pi_k \, \mathcal{N}(x_i \mid \mu_k, \sigma_k) \right)$$

```python
guesses = {
    'Bad guess\n(wrong centers, equal weights)': {
        'pis':    [1/3, 1/3, 1/3],
        'mus':    [-2.0, 0.0, 2.0],
        'sigmas': [1.0,  1.0, 1.0],
        'color':  'tomato',
    },
    'Good guess\n(close to truth)': {
        'pis':    pis_true,
        'mus':    mus_true,
        'sigmas': sigmas_true,
        'color':  'seagreen',
    },
}
```




```
Bad guess  LL: -1571.23
Good guess LL: -1166.43
Higher log-likelihood = better fit.
```

Estimating GMM using Maximum Likelihood

$$\mathcal{L}(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\sigma}) = -\sum_{i=1}^{N} \log \underbrace{\left(\sum_{k=1}^{K} \pi_k \, \mathcal{N}(x_i \mid \mu_k, \sigma_k) \right)}_{p(x_i)}$$

$$\pi_k = \frac{e^{\alpha_k}}{\sum_j e^{\alpha_j}} \quad \text{(softmax of free logits } \boldsymbol{\alpha}\text{)},$$

Symbol	Name	Meaning
π_k	**mixing weight**	How often this component is chosen. Must satisfy $\pi_k \geq 0$ and $\sum_k \pi_k = 1$.
μ_k	**mean**	Where the k-th Gaussian is centered.
σ_k	**standard deviation**	How wide the k-th Gaussian is.

We can use gradient descent

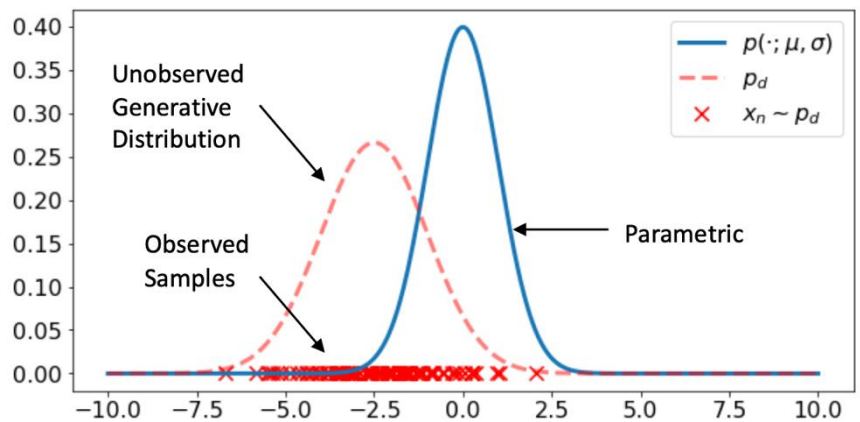
Estimating GMM using Maximum Likelihood – Theoretical Sol?

Gaussian Kernel Density Estimation

$$\{x_i | x_i \sim P_d\}_{i=1}^{N}$$

$$p(x;\mu,\sigma) = N(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

What is μ, σ ?



3- Minimizing Negative Log-Likelihood:

$$\underset{\mu,\sigma}{\operatorname{argmin}} \underbrace{\sum_{n=1}^{N} \frac{\log(2\pi\sigma^2)}{2} + \frac{(x_n-\mu)^2}{2\sigma^2}}_{\mathrm{L}} \longrightarrow \begin{cases} \dfrac{\partial L}{\partial \mu} = 0 \Rightarrow \mu_* = \dfrac{1}{N}\displaystyle\sum_{n=1}^{N} x_n \\ \dfrac{\partial L}{\partial \sigma} = 0 \Rightarrow \sigma_*^2 = \dfrac{1}{N}\displaystyle\sum_{n=1}^{N} (x_n - \mu^*)^2 \end{cases}$$

Estimating GMM using Maximum Likelihood – Theoretical Sol?

$$\ell(\mu,\sigma) = \log \prod_{i=1}^{N} \mathcal{N}(x_i \mid \mu,\sigma) = \sum_{i=1}^{N} \log \mathcal{N}(x_i \mid \mu,\sigma)$$

$$= \sum_{i=1}^{N} \left[-\log\sigma - \tfrac{1}{2}\log 2\pi - \frac{(x_i-\mu)^2}{2\sigma^2} \right]$$

$$p(x_i \mid \theta) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k,\sigma_k)$$

$$\ell(\theta) = \sum_{i=1}^{N} \log \underbrace{\left[\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k,\sigma_k) \right]}_{\text{sum inside the log}}$$

$$\frac{\partial \ell}{\partial \mu_k} = \sum_{i=1}^{N} \frac{\pi_k \mathcal{N}(x_i \mid \mu_k,\sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j,\sigma_j)} \cdot \frac{x_i-\mu_k}{\sigma_k^2} = 0$$

Estimating GMM using Maximum Likelihood – Theoretical Sol?

$$\ell(\mu,\sigma) = \log \prod_{i=1}^{N} \mathcal{N}(x_i \mid \mu,\sigma) = \sum_{i=1}^{N} \log \mathcal{N}(x_i \mid \mu,\sigma)$$

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Different parameters they are coupled together

$$\boxed{\frac{\partial \ell}{\partial \mu_k} = \sum_{i=1}^{N} \frac{\pi_k \mathcal{N}(x_i \mid \mu_k,\sigma_k)}{\sum_j \pi_j \mathcal{N}(x_i \mid \mu_j,\sigma_j)} \cdot \frac{x_i-\mu_k}{\sigma_k^2} = 0}$$

Estimating GMM using Maximum Likelihood – Theoretical Sol?

$$p(x_i \mid \theta) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k)$$

$$z_{ik} = \begin{cases} 1 & \text{if point } i \text{ was generated by component } k \\ 0 & \text{otherwise} \end{cases}$$

$$\ell(\theta) = \sum_{i=1}^{N} \log \underbrace{\left[\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k) \right]}_{\text{sum inside the log}}$$

$$\ell_{\text{complete}}(\theta) = \sum_{i=1}^{N} \sum_{k=1}^{K} z_{ik} \underbrace{[\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \sigma_k)]}_{\text{log is now INSIDE — easy!}}$$

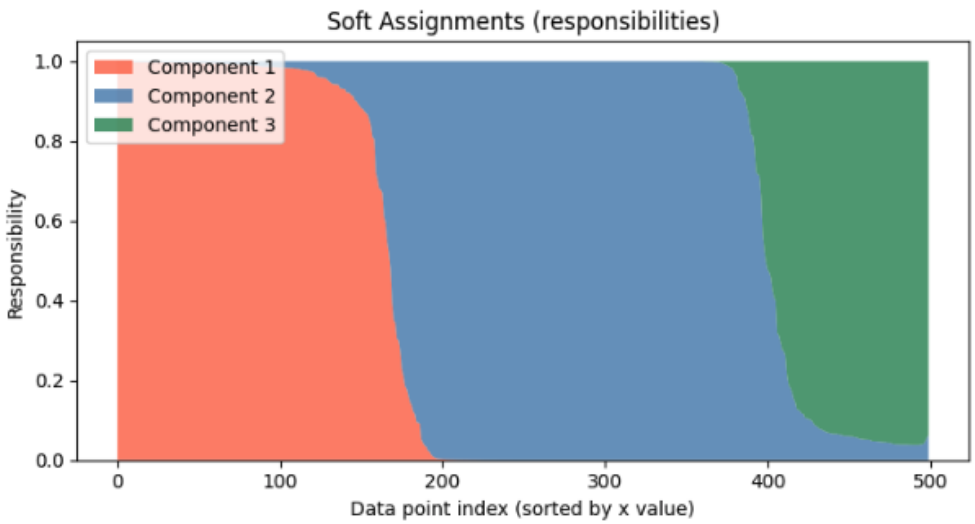
$$\ell_{\text{complete}}(\theta) = \underbrace{\sum_{k=1}^{K} \left(\sum_{i=1}^{N} z_{ik} \right) \log \pi_k}_{\text{depends only on } \pi} + \sum_{k=1}^{K} \underbrace{\sum_{i=1}^{N} z_{ik} \left[-\log \sigma_k - \tfrac{1}{2} \log 2\pi - \frac{(x_i - \mu_k)^2}{2\sigma_k^2} \right]}_{\text{depends only on } \mu_k, \sigma_k \text{ — independent across } k}$$

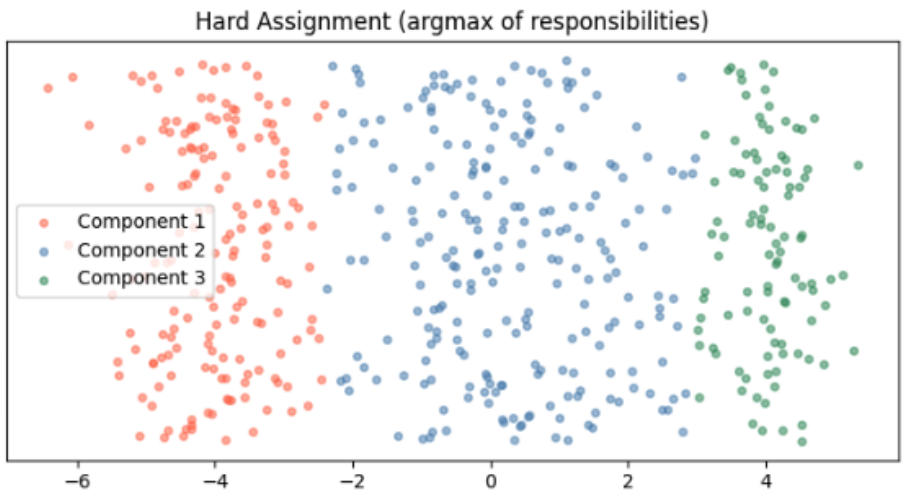
Once we introduce z, different parameters are decoupled

Estimating GMM using Maximum Likelihood – Theoretical Sol?

$$\ell_{\text{complete}}(\theta) = \underbrace{\sum_{k=1}^{K}\left(\sum_{i=1}^{N} z_{ik}\right)\log \pi_k}_{\text{depends only on } \boldsymbol{\pi}} + \underbrace{\sum_{k=1}^{K}\sum_{i=1}^{N} z_{ik}\left[-\log\sigma_k - \tfrac{1}{2}\log 2\pi - \frac{(x_i-\mu_k)^2}{2\sigma_k^2}\right]}_{\text{depends only on } \mu_k,\,\sigma_k \text{ — independent across } k}$$

Once we introduce z, different parameters are decoupled





Estimating GMM using Maximum Likelihood – EM Algorithm

E - Step

$$p(x_i \mid \theta) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k)$$

$$z_{ik} = \begin{cases} 1 & \text{if point } i \text{ was generated by component } k \\ 0 & \text{otherwise} \end{cases}$$

$$\ell(\theta) = \sum_{i=1}^{N} \log \underbrace{\left[\sum_{k=1}^{K} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k) \right]}_{\text{sum inside the log}}$$

$$\ell_{\text{complete}}(\theta) = \sum_{i=1}^{N} \sum_{k=1}^{K} z_{ik} \underbrace{[\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \sigma_k)]}_{\text{log is now INSIDE — easy!}}$$

z_{ik}: how much contribution does k^{th} gaussian represent to i^{th} data?

$$r_{ik} = \mathbb{E}[z_{ik} \mid x_i, \theta^{\text{old}}] = P(z_{ik} = 1 \mid x_i, \theta^{\text{old}})$$

$$P(x_i \mid z_i{=}k)\, P(z_i{=}k)$$

$$\boxed{r_{ik} = \frac{\pi_k^{\text{old}} \mathcal{N}(x_i \mid \mu_k^{\text{old}}, \sigma_k^{\text{old}})}{\sum_{j=1}^{K} \pi_j^{\text{old}} \mathcal{N}(x_i \mid \mu_j^{\text{old}}, \sigma_j^{\text{old}})}}$$

Estimating GMM using Maximum Likelihood – EM Algorithm

M - Step

$$\ell_{\text{complete}}(\theta) = \sum_{i=1}^{N}\sum_{k=1}^{K} \underbrace{z_{ik}}_{\text{unknown}} \left[\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \sigma_k)\right] \longrightarrow \sum_{i=1}^{N}\sum_{k=1}^{K} \underbrace{r_{ik}}_{\text{known}} \left[\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \sigma_k)\right]$$

z_{ik}: *how much contribution does* k^{th} *gaussian represent to* i^{th} *data*?

5a — Update π_k

$$\sum_k \pi_k = 1,\ \pi_k \geq 0 \qquad \frac{\partial}{\partial \pi_k}\left[\ell_{\text{complete}} - \lambda\left(\sum_k \pi_k - 1\right)\right] = \frac{N_k}{\pi_k} - \lambda = 0 \implies \pi_k = \frac{N_k}{\lambda}$$

$$\sum_k \pi_k = 1 \Rightarrow \lambda = \sum_k N_k = N.$$

$$\boxed{\pi_k^{\text{new}} = \frac{N_k}{N}, \qquad N_k = \sum_{i=1}^{N} r_{ik}}$$

Estimating GMM using Maximum Likelihood – EM Algorithm

M - Step

$$\ell_{\text{complete}}(\theta) = \sum_{i=1}^{N}\sum_{k=1}^{K} \underbrace{z_{ik}}_{\text{unknown}} \left[\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \sigma_k)\right] \longrightarrow \sum_{i=1}^{N}\sum_{k=1}^{K} \underbrace{r_{ik}}_{\text{known}} \left[\log \pi_k + \log \mathcal{N}(x_i \mid \mu_k, \sigma_k)\right]$$

z_{ik}: how much contribution does k^{th} gaussian represent to i^{th} data?

5b — Update μ_k

5c — Update σ_k

$$\sum_{i=1}^{N} r_{ik} \log \mathcal{N}(x_i \mid \mu_k, \sigma_k) = -\frac{1}{2\sigma_k^2}\sum_{i=1}^{N} r_{ik}(x_i - \mu_k)^2 + \text{const}$$

$$\frac{1}{\sigma_k^2}\sum_{i=1}^{N} r_{ik}(x_i - \mu_k) = 0 \implies \sum_i r_{ik} x_i = \mu_k N_k$$

$$-\frac{N_k}{\sigma_k} + \frac{1}{\sigma_k^3}\sum_{i=1}^{N} r_{ik}(x_i - \mu_k^{\text{new}})^2 = 0$$

$$\boxed{\mu_k^{\text{new}} = \frac{\sum_{i=1}^{N} r_{ik}\, x_i}{N_k}}$$

$$\boxed{\sigma_k^{\text{new}} = \sqrt{\frac{\sum_{i=1}^{N} r_{ik}(x_i - \mu_k^{\text{new}})^2}{N_k}}}$$

Estimating GMM using Maximum Likelihood – EM Algorithm

```
initialise  π, μ, σ  randomly

repeat:
    E-step:  r_ik  ←  Bayes' theorem  (Step 4)
    M-step:  N_k   ←  Σ_i r_ik
             π_k   ←  N_k / N
             μ_k   ←  Σ_i(r_ik · x_i) / N_k
             σ_k   ←  sqrt(Σ_i(r_ik · (x_i − μ_k)²) / N_k)
until  |ΔLL| < ε
```

```python
# — Gaussian PDF (manual, no scipy wrapper) ————————————
def gaussian_pdf(x, mu, sigma):
    """N(x | mu, sigma) computed directly from the formula."""
    coeff  = 1.0 / (sigma * np.sqrt(2 * np.pi))
    kernel = np.exp(-0.5 * ((x - mu) / sigma) ** 2)
    return coeff * kernel
```

```python
# — E-step: Bayes' theorem → responsibilities ————————————
def e_step(x, pis, mus, sigmas):
    """
    Compute r[i

    From Bayes:

    Returns r: array of shape (N, K).
    """
    N, K = len(x), len(pis)

    # Numerator: pi_k * N(x_i | mu_k, sigma_k)  shape (N, K)
    numer = np.column_stack([
        pis[k] * gaussian_pdf(x, mus[k], sigmas[k])
        for k in range(K)
    ])

    # Denominator: p(x_i) = sum over k of numerator row   shape (N, 1)
    denom = numer.sum(axis=1, keepdims=True)

    r = numer / denom          # soft assignments, each row sums to 1
    return r
```

$$r_{ik} = \frac{\pi_k^{\text{old}} \mathcal{N}(x_i \mid \mu_k^{\text{old}}, \sigma_k^{\text{old}})}{\sum_{j=1}^{K} \pi_j^{\text{old}} \mathcal{N}(x_i \mid \mu_j^{\text{old}}, \sigma_j^{\text{old}})}$$

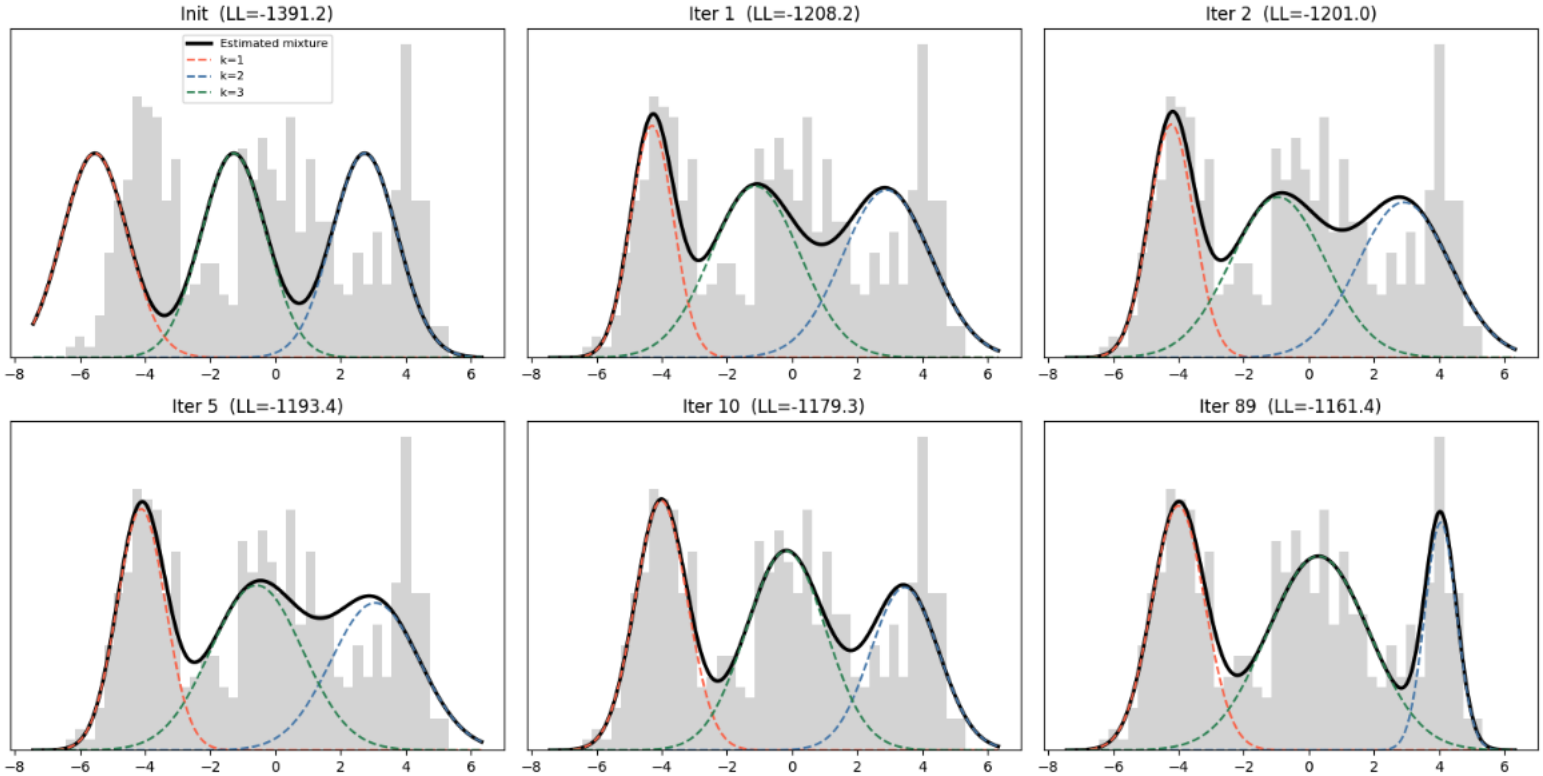
```python
# — M-step: closed-form updates from the Q-function derivatives ————————
def m_step(x, r):
    """
    Update parameters by maximising the expected complete-data log-likelihood.

    Derived formulas:
        N_k     = sum_i r_ik                        (effective count)
        pi_k    = N_k / N                           (from Lagrange multiplier)
        mu_k    = sum_i(r_ik * x_i) / N_k           (weighted mean)
        sigma_k = sqrt(sum_i(r_ik*(x_i-mu_k)^2) / N_k)   (weighted std)

    Returns new_pis, new_mus, new_sigmas.
    """
    N, K = r.shape

    # Effective count per component
    Nk = r.sum(axis=0)

    # pi update — from d/d(pi_k)[sum N_k log pi_k - lambda(sum pi_k
    new_pis = Nk / N

    # mu update — from d/d(mu_k)[sum r_ik * log N(x_i|mu_k,sigma_k)]
    new_mus = (r * x[:, None]).sum(axis=0) / Nk   # shape (K,)

    # sigma update — from d/d(sigma_k)[...] = 0
    residuals_sq = (x[:, None] - new_mus[None, :]) ** 2   # (N, K
    new_sigmas   = np.sqrt((r * residuals_sq).sum(axis=0) / Nk)

    return new_pis, new_mus, new_sigmas
```

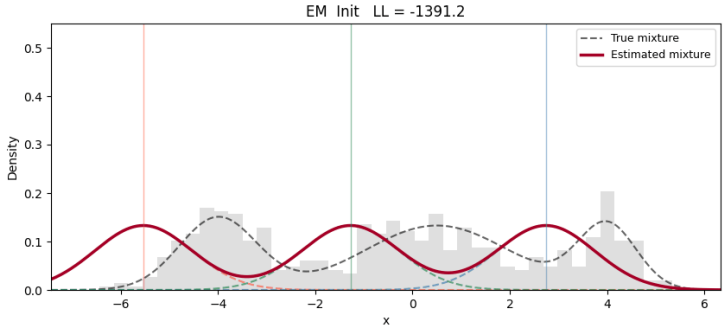
$$\pi_k^{\text{new}} = \frac{N_k}{N}, \qquad N_k = \sum_{i=1}^{N} r_{ik}$$

$$\mu_k^{\text{new}} = \frac{\sum_{i=1}^{N} r_{ik} x_i}{N_k}$$

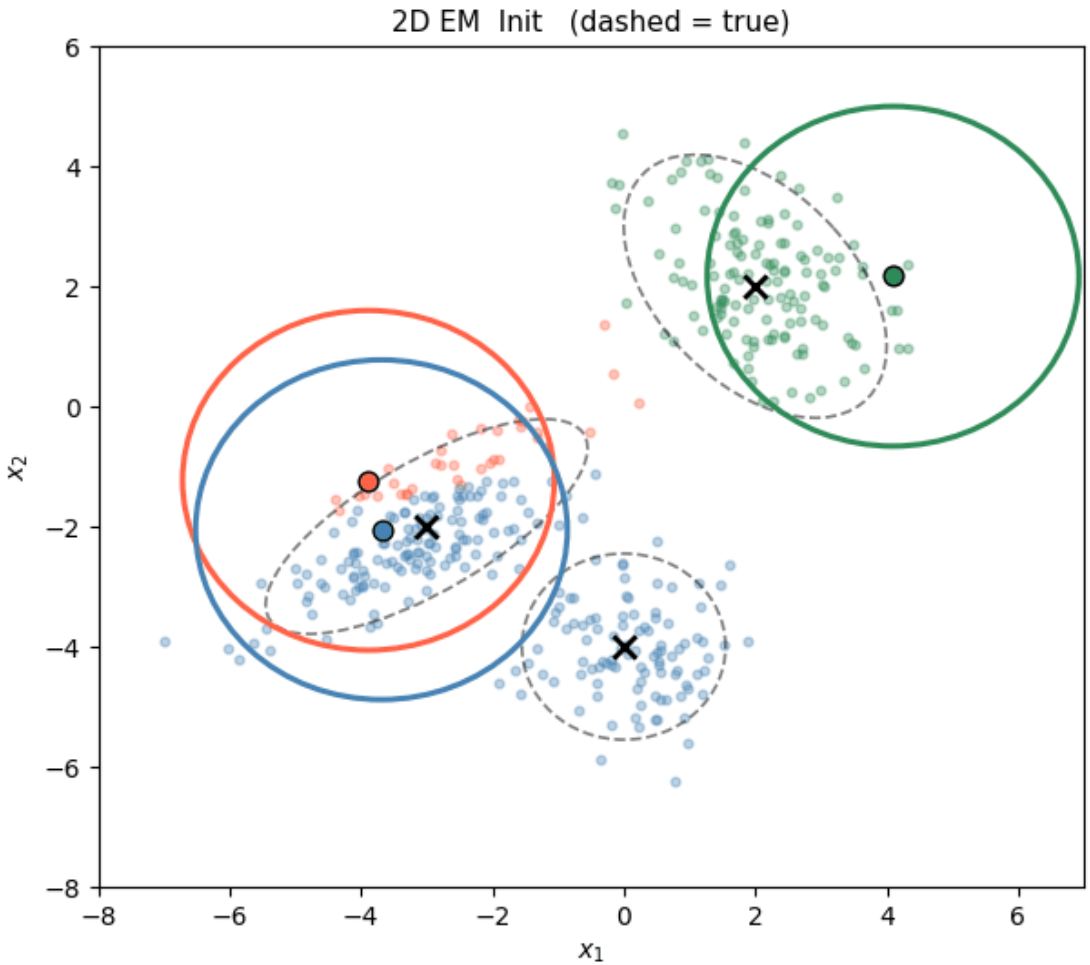
$$\sigma_k^{\text{new}} = \sqrt{\frac{\sum_{i=1}^{N} r_{ik}(x_i - \mu_k^{\text{new}})^2}{N_k}}$$

Estimating GMM using Maximum Likelihood – EM Algorithm





Estimating GMM using Maximum Likelihood – EM Algorithm



Estimating GMM using Maximum Likelihood – Gradient Descent

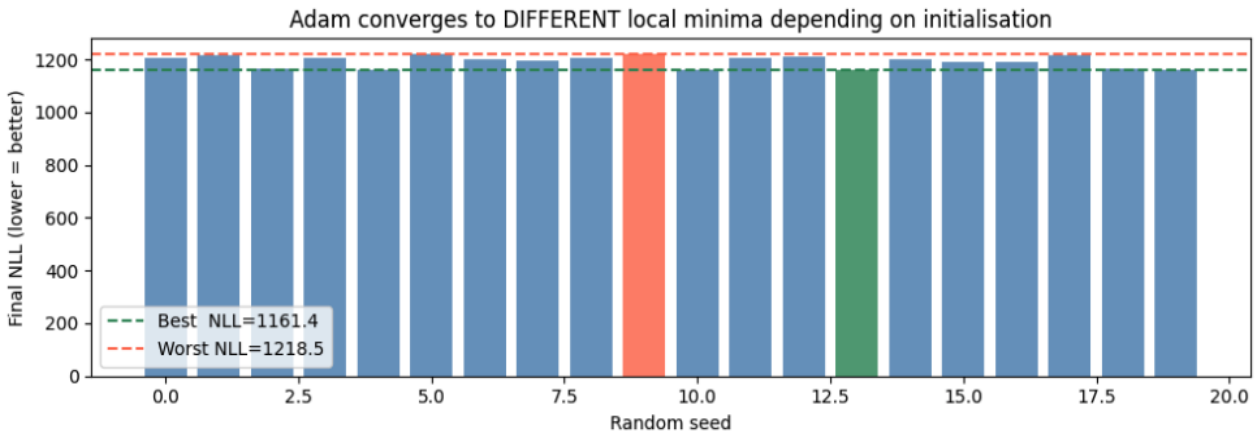
$$\mathcal{L}(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\sigma}) = -\sum_{i=1}^{N} \log \underbrace{\left(\sum_{k=1}^{K} \pi_k \, \mathcal{N}(x_i \mid \mu_k, \sigma_k) \right)}_{p(x_i)} \qquad \pi_k = \frac{e^{\alpha_k}}{\sum_j e^{\alpha_j}} \quad \text{(softmax of free logits } \boldsymbol{\alpha}\text{)},$$

Symbol	Name	Meaning
π_k	**mixing weight**	How often this component is chosen. Must satisfy $\pi_k \geq 0$ and $\sum_k \pi_k = 1$.
μ_k	**mean**	Where the k-th Gaussian is centered.
σ_k	**standard deviation**	How wide the k-th Gaussian is.

```python
x_torch = torch.tensor(x_samples, dtype=torch.float32)
K = 3

def gmm_nll_torch(x, pi_logits, mus, log_sigmas):
    """Negative log-likelihood of a GMM (PyTorch, differentiable)."""
    pis    = F.softmax(pi_logits, dim=0)
    sigmas = torch.exp(log_sigmas)
    log_components = (
        torch.log(pis)[None, :]
        - log_sigmas[None, :]
        - 0.5 * np.log(2 * np.pi)
        - 0.5 * ((x[:, None] - mus[None, :]) / sigmas[None, :]) ** 2
    )
    return -torch.logsumexp(log_components, dim=1).sum()

def run_adam(seed, n_iters=500, lr=0.05):
    torch.manual_seed(seed)
    pi_logits  = torch.randn(K, requires_grad=True)
    mus_t      = torch.randn(K, requires_grad=True)
    log_sigmas = torch.zeros(K, requires_grad=True)
    opt = torch.optim.Adam([pi_logits, mus_t, log_sigmas], lr=lr)
    for _ in range(n_iters):
        opt.zero_grad()
        gmm_nll_torch(x_torch, pi_logits, mus_t, log_sigmas).backward()
        opt.step()
    with torch.no_grad():
        return (F.softmax(pi_logits, dim=0).numpy().copy(),
                mus_t.numpy().copy(),
                torch.exp(log_sigmas).numpy().copy(),
                gmm_nll_torch(x_torch, pi_logits, mus_t, log_sigmas).item())
```



Estimating GMM using Maximum Likelihood – Gradient Descent

$$\mathcal{L}(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\sigma}) = -\sum_{i=1}^{N} \log \underbrace{\left(\sum_{k=1}^{K} \pi_k \, \mathcal{N}(x_i \mid \mu_k, \sigma_k) \right)}_{p(x_i)} \qquad \pi_k = \frac{e^{\alpha_k}}{\sum_j e^{\alpha_j}} \quad \text{(softmax of free logits } \boldsymbol{\alpha}\text{)},$$

Symbol	Name	Meaning
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        - 0.5 * ((x[:, None] - mus[None, :]) / sigmas[None, :]) ** 2
    )
    return -torch.logsumexp(log_components, dim=1).sum()

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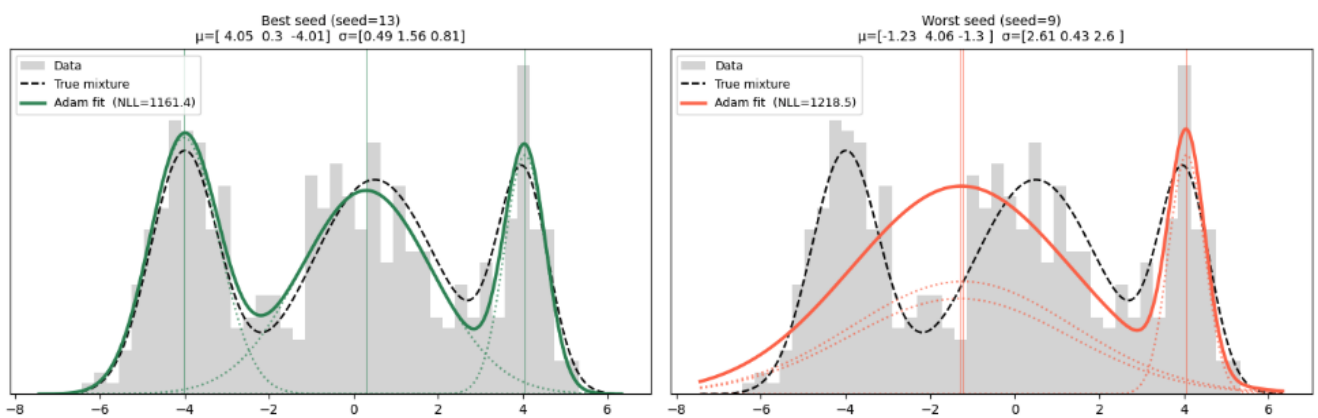
    mus_t      = torch.randn(K, requires_grad=True)
    log_sigmas = torch.zeros(K, requires_grad=True)
    opt = torch.optim.Adam([pi_logits, mus_t, log_sigmas], lr=lr)
    for _ in range(n_iters):
        opt.zero_grad()
        gmm_nll_torch(x_torch, pi_logits, mus_t, log_sigmas).backward()
        opt.step()
    with torch.no_grad():
        return (F.softmax(pi_logits, dim=0).numpy().copy(),
                mus_t.numpy().copy(),
                torch.exp(log_sigmas).numpy().copy(),
                gmm_nll_torch(x_torch, pi_logits, mus_t, log_sigmas).item())
```



QA